

A Digital Rights, Safety & Wellbeing Support System

System Blueprint & Technical Architecture — Website, AI Chatbot & Anonymous WhatsApp Helpline

Prepared as a build-ready specification for engineering teams and AI coding assistants (e.g. GitHub Copilot). Grounded in the working areas of Amnesty International and the standards of the United Nations (UN), World Health Organization (WHO), and recognised digital-rights and child-safeguarding organisations. Version 1.0.

Field	Detail
Product name (working title)	Amani Line — "Amani" (Swahili/Twi-adjacent for peace) Rights & Wellbeing Support Network
Channels	Public website · Web chat widget · Anonymous WhatsApp number · (Phase 2) SMS/USSD fallback
Core purpose	Give people confidential, non-judgemental information, verified referrals and links to real human help across 9 rights & wellbeing themes
Primary users	Youth activists, protesters, students, journalists, human rights defenders, LGBTQ+ & gender-rights advocates, people in mental-health distress, people targeted by online harm, and men seeking peer support
Geographic scope	Ghana-first (local laws, local referral bodies, local languages), extensible to any country via a configurable referral database
Guiding frameworks	Amnesty International thematic priorities · UN Guiding Principles on Business & Human Rights · WHO mental-health & safe-messaging guidance · UNHRC digital rights resolutions · GDPR & Ghana Data Protection Act, 2012 (Act 843)

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1. Vision & Problem Statement

Young people in Ghana and beyond who engage in protest, activism, digital expression, gender advocacy, environmental justice, mental-health struggles or governance participation often face the same three gaps: they don't know their rights, they don't know who to trust for information, and they don't know a safe, anonymous way to ask for help. Misinformation, phishing and online harassment compound the risk. This system closes that gap with a single, trustworthy front door — a website and an anonymous WhatsApp line — backed by an AI assistant that informs, reassures and refers, but never replaces professional or emergency services.

The system deliberately follows Amnesty International's own working method: research the facts, protect the individual, and mobilise the appropriate institutions — rather than tell people what action to personally take. It also names and pushes back on harms Amnesty and partner bodies track closely in Ghana and the wider region, including violent responses to protest, attacks on civic space, digital surveillance, and harmful traditional practices such as witchcraft accusations against women and older persons.

2. Design Principles & Ethical Guardrails

- **Inform, don't instruct:** the AI explains rights, laws and options, then refers to a real body or hotline — it never tells a user to personally carry out a specific risky action (e.g. "go to the protest", "confront your abuser").
- **Anonymous by default:** no phone-number-to-identity linking is stored beyond what WhatsApp itself requires operationally; conversations are pseudonymous unless the user actively opts to share identifying details to be connected to a named contact.
- **Do no harm / safe messaging:** for suicide, self-harm and gender-based violence topics, the assistant follows WHO/International Association for Suicide Prevention (IASP) safe-messaging principles — no method or means detail, calm and hopeful tone, always paired with a live helpline or professional referral.
- **Human-in-the-loop escalation:** any message indicating imminent danger (self-harm, threat of violence, arrest in progress, trafficking) is flagged for priority human moderator handoff, never left to the bot alone.
- **Neutrality on contested politics:** the system presents rights, laws and verified facts, not partisan opinions; it never tells a user which political position to hold, only what their legal rights and safety options are.
- **Global & local law together:** every answer that touches rights pairs the Ghanaian legal position with the applicable international instrument (ICCPR, African Charter, UNCRC, CEDAW, etc.).
- **Radical transparency about limits:** the bot always discloses that it is an AI, that it is not a lawyer, doctor or counsellor, and that it can connect the user to one.
- **Privacy & consent first:** data minimisation, encryption in transit and at rest, a visible "delete my chat history" control, and no advertising or data resale — modelled on Amnesty's own Human Rights Friendly design commitments and human-rights-based approach to technology.
- **Inclusive & intersectional:** content and referral paths are reviewed for gender, disability, LGBTQ+, ethnic and religious inclusiveness, including a dedicated, non-judgemental peer-support pathway for men and boys ("Men's Circle") alongside the feminist/gender-justice pathway — not positioned as opposing tracks, but as two doors into the same support ecosystem.

3. The Nine Support Pillars

Mapped directly from the research-group topics, plus two additions you specified (men's peer support and anti-witchcraft-accusation work, which sits inside Amnesty's older-persons & women's rights programme).

Pillar	What the system actually does
1. Protest Rights & Civic Freedom	Know-your-rights explainers (arrest, assembly, police conduct), legal aid contacts, real-time "what to do if detained" referral (not instructions to act) — sourced from Amnesty, CHRAJ, Ghana Police Service standing orders.
2. Activism & Youth-Led Movements	Toolkits on organising safely, digital security basics for organisers, directory of accredited civil-society networks (e.g. CDD-Ghana, WACSI, STAR-Ghana).
3. Suicide Prevention & Youth Mental Health	WHO-aligned safe-messaging responses, always paired with a live crisis line (Ghana Mental Health Authority helpline, Mind Freedom Ghana), professional referral, never method/means detail.
4. Climate Justice & Environmental Rights	Verified climate-rights explainers, links to EPA Ghana, 350Africa, Amnesty's climate-justice programme, reporting channel for environmental harms (e.g. galamsey-linked abuses).
5. Digital Rights, Online Safety & Freedom of Expression	Phishing/scam-link checker, digital-security how-tos, cyberbullying and doxxing referral, Ghana Cybersecurity Authority & Data Protection Commission contacts, Access Now Digital Security Helpline.
6. Youth Participation in Governance	Civic-education content (NCCE-aligned), voter and local-government participation guides, links to Ghana Youth Authority & parliamentary petition channels.
7. Human Rights Defenders' Safety & Security	Rapid-referral to Front Line Defenders, Amnesty Urgent Action network, West Africa Human Rights Defenders Network, physical/digital security check-lists.
8. Gender, Intersectionality & Feminist Activism	GBV and harassment referral (DOVVSU, Ark Foundation, WiLDAF), reporting pathways, education on CEDAW/Maputo Protocol rights.
9. Men's Circle (Positive Masculinity Support)	Non-judgemental peer-support entry for men facing violence, addiction, fatherhood strain, or mental-health stigma; referral to Ghana Mental Health Authority, MenEngage Africa, licensed counsellors — framed as complementary to, not competing with, gender-justice work.

A cross-cutting tenth theme — **combating witchcraft accusations and "witch camp" abuses** — is woven through pillars 1, 8 and 9 as a standing content module, reflecting Amnesty Ghana's active campaigning on this issue.

4. User Journey — How a Real Person Uses the System

- **Discovery:** user finds the WhatsApp number or website via social media, campus posters, or an Amnesty/partner-organisation referral.
- **First contact (anonymous):** user texts the WhatsApp number or opens the web chat. No sign-up, no name required. Bot greets them, states clearly it is an AI, explains confidentiality, and asks (via quick-reply buttons) which of the 9 pillars is closest to their need — or lets them type freely.
- **Triage:** a lightweight safety-classifier scans the message for urgent-risk signals (self-harm, imminent violence, minors at risk, active detention). Urgent cases skip straight to a human-handoff queue with a locally relevant emergency number surfaced immediately.
- **Information + referral:** for non-urgent queries, the AI retrieves verified information (see Section 8) and gives a short, plain-language answer, then offers 1–3 concrete next steps: a verified link, a hotline, an organisation's contact, or "talk to a real person" handoff.
- **Localisation:** if the user shares a region (optional, never demanded), the bot narrows referrals to that region's police unit, hospital, CSO chapter, or legal-aid office.
- **Optional identification:** only if the user chooses to be connected to a named case-worker (e.g. for legal aid follow-up) does the system ask for contact details, under an explicit, separate consent step, stored encrypted and deletable.
- **Exit & follow-up:** user can end the chat anytime; a "forget this conversation" command permanently deletes the anonymous thread. Optional satisfaction/feedback survey (2 taps) improves the service without collecting identity.

5. System Architecture Overview

The system is organised in five layers:

Layer	Components
1. Channel layer	Public website (React) · Web chat widget · WhatsApp Cloud API number · (Phase 2) SMS/USSD gateway
2. Conversation & orchestration layer	API gateway → Conversation Orchestrator (intent routing, safety triage, session state) → Human-handoff queue for moderators/partner-org staff
3. AI / Knowledge layer	LLM (guarded system prompt + tool-calling) + Retrieval-Augmented Generation (RAG) over a curated, versioned knowledge base; separate lightweight safety-classifier model for risk triage
4. Trust & safety services	URL/phishing scanner (Safe Browsing/VirusTotal APIs) · Content moderation filter · Rate-limiting & abuse detection · Human moderator dashboard
5. Data & integration layer	Encrypted conversation store (short-retention, anonymised) · Referral-directory database (organisations, hotlines, geocoded) · Content-source registry (allow-listed publishers) · Admin CMS for partner organisations to keep referral data current

Simplified request flow

User message (WhatsApp/Web) → API Gateway → Safety Triage Classifier → *if urgent*: instant crisis-resource message + human moderator alert; *if routine*: Conversation Orchestrator → RAG retrieval from Knowledge Base + Referral Directory → LLM composes a grounded, referral-first reply → Trust & Safety filter (checks for banned advice patterns, verifies any link against the phishing scanner) → reply sent to user → anonymised interaction log stored for quality review only.

6. Recommended Technology Stack

Layer	Recommended choice
Website / front-end	React + TypeScript (Next.js), Tailwind CSS design system in Amnesty colours, WCAG 2.2 AA accessibility, multilingual (English, Twi, Ga, Ewe, Hausa) via i18next
Web chat widget	Embedded React chat component talking to the same Conversation API used by WhatsApp, so logic lives once and renders twice
Messaging channel	WhatsApp Business Platform (Cloud API) via Meta directly or an approved Business Solution Provider (e.g. Twilio, 360dialog, ChatDaddy); dedicated, published "Support Line" number with a verified green-tick business profile clearly stating it is an anonymous support & information service, not an emergency line
Backend / orchestration	Node.js (NestJS) or Python (FastAPI) microservices; message broker (e.g. Redis Streams or RabbitMQ) between channel webhooks and the orchestrator
AI / LLM layer	Claude or GPT-class model via API with a strict, versioned system prompt + tool use; RAG via a vector database (pgvector on Postgres, or a managed vector store) over the curated knowledge base; a smaller/faster classifier model (or fine-tuned lightweight model) purely for risk triage, kept separate from the conversational model for auditability
Knowledge base & CMS	Headless CMS (e.g. Strapi) so legal officers/partner NGOs can edit rights information and referral contacts without touching code; every fact entry stores its source URL, publisher, and last-verified date
Referral directory	PostgreSQL with PostGIS extension for location-aware matching of hotlines/organisations to a user's stated region
Link/phishing safety	Google Safe Browsing API + VirusTotal API + PhishTank feed, aggregated behind a single internal "URL reputation" microservice
Security	OAuth2/JWT for admin access, Signal-protocol-style E2E encryption for the web chat transport, TLS 1.3 everywhere, secrets in a managed vault (e.g. AWS Secrets Manager / HashiCorp Vault)
Infrastructure	Cloud-hosted (AWS, Azure, or Google Cloud) with data residency consideration for Ghana/EU; containerised services (Docker + Kubernetes) for independent scaling of the AI, safety and channel layers
Observability	Centralised logging with PII redaction, anonymised analytics dashboard (Metabase/Looker) tracking topic volume, escalation rate and referral click-through — never individual identities

7. AI / Chatbot Behaviour Specification

The assistant's system prompt and tool-access should encode the following behaviour rules as hard constraints, independent of what a user asks it to do:

- Always identify itself as an AI assistant, never impersonate a human counsellor or lawyer.
- Answer only from the verified knowledge base and referral directory (RAG-grounded) — it does not freely generate legal, medical or safety claims from unguided model memory.
- For every substantive answer, name at least one concrete referral (organisation, hotline, or official source) the user can contact next.
- Refuse to give operational instructions for risky personal action (e.g. how to evade police, how to confront an abuser, self-harm methods); instead it explains rights/options and refers to a body equipped to help.
- On any signal of imminent risk to life (self-harm intent, active violence, a minor in danger), it immediately surfaces the relevant crisis line and flags the conversation for human moderator review, without waiting for further prompting.
- Uses a warm, calm, non-judgemental tone; avoids alarmist or graphic language, consistent with WHO safe-messaging guidance.
- Declines to take or endorse a partisan political position; presents rights and facts, not opinions on contested policy questions.
- Supports short free-text queries and a structured menu (quick-reply buttons on WhatsApp) so users who don't know what to type can still navigate the 9 pillars.
- Logs a machine-readable "outcome tag" per conversation (e.g. information-given, referred-to-hotline, escalated-to-human) for anonymised service reporting — never logs raw personal identifiers unless the user explicitly opted into being contacted.

8. Verified-Information & Fact-Sourcing Pipeline

Rather than letting the AI browse the open internet live for every answer (which risks pulling in unverified or manipulated content), the recommended design uses a **curated-then-augmented** pipeline:

- **Allow-listed source registry:** a maintained list of trusted publishers — Amnesty International, UN OHCHR, WHO, UNICEF, UNDP, Ghana's CHRAJ, Ghana Police Service, Ghana Health Service, Mental Health Authority, Ghana Data Protection Commission, reputable Ghanaian outlets (GhanaWeb, Joy News, MyJoyOnline, Graphic) and recognised legal-aid bodies.
- **Scheduled ingestion:** a background job periodically fetches updates from these sources, extracts the relevant facts, and stores them with source URL, publish date and a "last verified" timestamp — this becomes the RAG knowledge base the chatbot draws from.
- **Human review gate:** a small editorial/legal team (partner NGO staff or volunteer lawyers) reviews new or changed entries before they go live, especially for legal-rights content, mirroring how Amnesty's own research is vetted before publication.
- **Live web-search fallback (optional, rate-limited):** for fast-moving news (e.g. a new law, a breaking protest event), the system can call a live search API, but results are shown as "unverified — for your awareness" with a link, clearly separated from the vetted knowledge-base answers, until reviewed and promoted into the permanent knowledge base.
- **Freshness indicator:** every answer footnotes "last verified on [date]" so users can judge currency.

9. Anonymous WhatsApp Helpline Design

- **Number & onboarding:** a single published WhatsApp Business number; first message triggers an automated welcome explaining what the service is, that it is anonymous, that it is not an emergency service, and offering a menu of the 9 pillars plus "talk to a human" and "emergency numbers" quick replies.
- **Anonymity model:** WhatsApp requires a phone number to deliver messages, so true anonymity from Meta/the telecom is not possible — the system's commitment is that *the platform itself* does not store the user's name, does not link the number to any other identity system, encrypts the number at rest, and purges numbers from analytics after a short retention window (e.g. 30–90 days) unless the user has opted into ongoing case support.
- **Session & language:** language auto-detected or chosen at onboarding (English, Twi, Ga, Ewe, Hausa); session state kept only long enough to hold a conversation, not built into a persistent CRM profile.
- **Escalation path:** a "talk to a human" quick reply hands off to a rotating roster of trained volunteers/partner-org staff via a moderator dashboard, with the AI conversation history shown to the human agent only after user consent (or automatically in a flagged imminent-risk case, disclosed to the user as standard practice).
- **Rate & abuse handling:** automated throttling and basic spam/abuse detection so the line cannot be flooded or used to harass staff; clearly stated acceptable-use notice.
- **Cost model:** WhatsApp Business Platform (Cloud API) charges per conversation/message and is typically accessed through a Meta-approved Business Solution Provider (BSP) such as Twilio, 360dialog or a regional provider — budget for BSP platform fees plus Meta's per-message/category charges, which should be modelled against expected monthly volume before committing to a provider.

10. Link & Phishing Safety Module

Supports pillar 5 (Digital Rights & Online Safety) by letting a user paste a suspicious link, phone number, or screenshot text and get a safety verdict:

- URLs are checked against Google Safe Browsing and VirusTotal reputation APIs, plus a locally maintained block-list of known Ghanaian scam/phishing campaigns.
- The bot returns a plain-language verdict ("this link shows signs of phishing — do not enter your details") plus a short explanation of the red flags, never technical jargon.
- For confirmed scams targeting a wide public (e.g. a fake mobile-money promotion), the system can flag the case to the Ghana Cybersecurity Authority's incident-reporting channel with the user's consent.
- The module never itself "fixes" a compromised account; it always refers the user to the relevant provider's official recovery process or the Cybersecurity Authority.

11. Localised Referral Directory (Sample Seed Data — Ghana)

The directory is a living, editable database (not hard-coded), so the exact contact details below are illustrative seed entries for the build team to verify and load at launch — always confirm current numbers directly with each organisation before go-live.

Category	Example bodies to onboard
Human rights / civic space	Commission on Human Rights & Administrative Justice (CHRAJ); Amnesty International Ghana; CDD-Ghana
Mental health & suicide prevention	Ghana Mental Health Authority; Mind Freedom Ghana; a licensed tele-counselling partner
Gender-based violence	Domestic Violence & Victim Support Unit (DOVVSU), Ghana Police Service; Ark Foundation Ghana; WILDAF Ghana
Digital rights & cyber safety	Ghana Cybersecurity Authority incident-reporting channel; Media Foundation for West Africa (MFWA); Data Protection Commission
Environmental / climate justice	Environmental Protection Agency (EPA) Ghana; A Rocha Ghana; 350Africa
Governance & civic participation	National Commission for Civic Education (NCCE); Electoral Commission of Ghana
Human rights defenders	Front Line Defenders (international); Amnesty International's Urgent Action Network
Men's peer support	MenEngage Africa; a licensed counselling partner offering male-focused sessions
International reference bodies	UN Office of the High Commissioner for Human Rights (OHCHR); WHO; UNICEF; Amnesty International (global)

Directory schema fields: organisation name, category tag(s), region/coverage area, phone/WhatsApp/email/website, operating hours, languages supported, last-verified date, and an internal "trust tier" (official government body / accredited NGO / community volunteer network).

12. Privacy, Security & End-to-End Encryption

- **Transport encryption:** WhatsApp messages already travel over Meta's end-to-end encryption for user-to-business chat where supported; the website chat widget uses TLS 1.3 plus application-layer encryption of message payloads at rest.
- **Data minimisation:** collect only what is operationally necessary (message content, chosen topic, optional region); never require real names, national ID, or exact addresses.
- **Anonymisation & retention limits:** conversation logs are pseudonymised (hashed session IDs, not phone numbers) for analytics, with a short default retention period (e.g. 90 days) after which raw content is purged; only aggregated, de-identified statistics are kept long-term.
- **Right to be forgotten:** a user-triggered "delete my chat" command purges their thread immediately, in line with the Ghana Data Protection Act, 2012 (Act 843) and GDPR-equivalent principles for any EU-based users or funders.
- **Access control:** role-based access for moderators/case-workers, full audit logging of any human access to a conversation, and mandatory data-protection and safeguarding training for all staff/volunteers with access.
- **Consent-gated identification:** a user is only ever identified if they explicitly choose to be connected to a named case-worker for follow-up (e.g. legal aid); this is a distinct, clearly explained opt-in step, never a default.
- **Independent security review:** a penetration test and a data-protection impact assessment (DPIA) before public launch, given the sensitivity of the subject matter.

13. Brand & UI/UX Guidelines (Amnesty-Aligned)

Amnesty International's global identity uses **Amnesty Yellow** as its primary colour — used sparingly, as an "alert" or call-to-action colour rather than a decorative fill — paired with black and white for contrast and legibility, plus a restrained grey for supporting graphics. This system should borrow that logic rather than the literal logo (the Amnesty candle mark itself is protected and should not be used unless in a formal, permitted partnership).

Element	Value	Usage
Amnesty Yellow	#FFED00	Primary accent — buttons, highlights, urgent notices. Use sparingly.
Black	#111111	Primary text, headers, high-contrast backgrounds.
White	#FFFFFF	Dominant background for readability and calm tone.
Neutral grey	#EDED / #555555	Supporting UI chrome, secondary text, dividers.

- Plain, calm, accessible language (target reading level: lower-secondary school) in every UI string, in line with the tone described in Section 7.
- WCAG 2.2 AA colour-contrast and screen-reader support throughout the website and chat widget.
- A persistent, one-tap "Exit & clear history" control on every screen, and a visible "quick exit" button that instantly redirects to an innocuous page (useful for users in unsafe environments).
- Multilingual toggle (English, Twi, Ga, Ewe, Hausa at minimum) with content professionally translated, not machine-translated only, for legal/safety material.
- Mobile-first layout — most users will arrive via WhatsApp or a phone browser on modest data plans, so the website should be lightweight and fast on 3G.

14. Data Model — Core Entities

Entity	Key fields
Conversation	session_id (hashed), channel, language, start_ts, end_ts, outcome_tag, risk_flag
Message	conversation_id, sender_type (user/bot/human), content, timestamp, redacted_pii_flag
KnowledgeEntry	topic_pillar, title, body, source_url, publisher, last_verified_date, review_status
ReferralOrg	name, category, region, contact_channels, hours, languages, trust_tier, last_verified_date
URLCheckLog	hashed_url, verdict, reputation_source, timestamp
ModeratorAction	conversation_id, moderator_id (internal), action_type, timestamp, notes

15. Build Roadmap

Phase	Deliverables
Phase 0 — Foundation (4–6 weeks)	Finalise partner MOUs (Amnesty Ghana, CHRAJ, Mental Health Authority, etc.), assemble editorial/legal review team, set up cloud infrastructure, draft & legally review all knowledge-base content and the AI system prompt.
Phase 1 — MVP (6–10 weeks)	Website with static rights information for all 9 pillars; WhatsApp number live with rule-based menu + AI Q&A grounded in the initial knowledge base; manual human-handoff via a shared inbox; basic referral directory for Ghana.
Phase 2 — Core platform (8–12 weeks)	Full RAG pipeline with scheduled source ingestion; phishing/link-checker module; moderator dashboard with risk-triage queue; multilingual support; anonymised analytics dashboard.
Phase 3 — Scale & harden (ongoing)	Independent security audit & DPIA; SMS/USSD fallback for feature phones; regional expansion of the referral directory beyond Ghana; volunteer/partner case-worker network; periodic red-teaming of the AI's crisis-handling behaviour.

16. Governance, Safeguarding & Partnerships

- An advisory board including a human-rights lawyer, a licensed mental-health professional, a child-safeguarding specialist and a digital-security expert reviews the knowledge base and AI behaviour quarterly.
- Formal partnership agreements (not just informational links) with at least the core referral bodies — CHRAJ, Ghana Mental Health Authority, DOVVSU — so escalations reach a real, accountable contact point.
- A published safeguarding policy covering how the system handles disclosures involving minors, which is treated with extra caution and, where legally required, mandatory reporting obligations are followed transparently.
- Transparent public reporting (annual, anonymised) on usage volume, top topics, and outcomes, mirroring the transparency Amnesty itself expects of institutions.

17. Risks & Mitigations

Risk	Mitigation
AI gives incorrect legal/medical information	Strict RAG-grounding to a lawyer/clinician-reviewed knowledge base; the AI is barred from freely generating uncited legal or medical claims.
User in acute crisis is not handled fast enough	Dedicated, low-latency safety-classifier that runs before the conversational model, with an immediate crisis-resource message + human alert.
De-anonymisation risk via WhatsApp number	Number encrypted at rest, no cross-linking to other identity systems, short retention window, no marketing/analytics use of the raw number.
System used to spread the very misinformation it fights	Allow-listed source registry, human editorial review gate before any new fact goes live, visible "last verified" dating.
Government or third-party pressure to disclose user data	Data minimisation reduces what exists to disclose; publish a clear transparency/data-request policy in advance, consistent with Amnesty's own advocacy for protecting sources and defenders.
Over-reliance on the bot instead of real services	Every answer is referral-first by design; the bot is positioned publicly as a bridge to help, not a substitute for professional or emergency services.

18. Brief for GitHub Copilot / Build Agent

Paste the block below as the opening task description in your Copilot / coding-agent session to scaffold the repository in line with this blueprint.

Build a monorepo for "Amani Line" — a rights & wellbeing support platform — with:

1. /apps/web — Next.js + TypeScript + Tailwind website with pages for each of the 9 support pillars, an embedded chat widget component, and an accessible (WCAG 2.2 AA) design using colours #FFED00 (accent), #111111 (text/dark), #FFFFFF (background), #EDEDED (secondary).

2. /apps/whatsapp-gateway — Node.js (NestJS) service implementing a WhatsApp Cloud API webhook receiver and sender, forwarding normalised messages to the orchestrator over an internal message queue (Redis Streams).

3. /apps/orchestrator — FastAPI (Python) service that: (a) runs an incoming message through a safety-triage classifier, (b) if urgent, immediately returns a crisis-resource template and pushes a moderator alert; (c) if routine, performs RAG retrieval against a pgvector-backed knowledge base and a PostGIS-backed referral directory, then calls the LLM with a versioned system prompt to produce a grounded, referral-first response.

4. /apps/moderator-dashboard — React admin app for human moderators/partner-org staff: risk-triage queue, conversation handoff view (with consent-gated identity reveal), and CMS screens for editing KnowledgeEntry and ReferralOrg records.

5. /services/url-safety — a small microservice wrapping Google Safe Browsing + VirusTotal lookups behind one internal endpoint, used by the orchestrator and exposed as a "check this link" bot command.

6. /packages/shared — shared TypeScript/Python types for Conversation, Message, KnowledgeEntry, ReferralOrg, URLCheckLog and ModeratorAction, matching the data model in Section 14 of the attached blueprint.

Non-negotiable behaviour constraints for the orchestrator's LLM calls (encode directly in the system prompt): identify as an AI; answer only from retrieved, cited knowledge-base content; always include at least one concrete referral; never produce operational instructions for risky personal action; apply WHO safe-messaging rules to any suicide/self-harm-adjacent content and always pair it with a live crisis line; remain neutral on partisan political questions; flag imminent-risk conversations for human review automatically.

Use environment-based config for API keys (LLM provider, WhatsApp Cloud API, Safe Browsing, VirusTotal); do not commit secrets. Include Docker Compose for local development and a basic CI pipeline (lint, type-check, test) for each app.

19. Reference Standards & Sources

- Amnesty International — global thematic priorities (protest & civic space, digital rights, gender justice, human rights defenders, climate justice) and Amnesty Global Identity/Brand Toolkit (colour system).
- United Nations Office of the High Commissioner for Human Rights (OHCHR) — International Covenant on Civil and Political Rights (ICCPR); Human Rights Council resolutions on digital rights and the right to peaceful assembly.
- World Health Organization (WHO) & International Association for Suicide Prevention (IASP) — safe-messaging guidelines for suicide-and-self-harm-related content.
- UN Convention on the Rights of the Child (UNCRC) and CEDAW / Maputo Protocol — for child-safeguarding and gender-rights content standards.
- Ghana Data Protection Act, 2012 (Act 843), and the EU General Data Protection Regulation (GDPR) as a comparative privacy-by-design benchmark.
- Meta WhatsApp Business Platform (Cloud API) documentation — current architecture and per-message/category pricing model.
- Google Safe Browsing API, VirusTotal API and PhishTank — for the link/phishing safety module.
- Front Line Defenders and Access Now — models for human-rights-defender emergency referral and a digital-security helpline.

This document is a planning and engineering blueprint. It is not a substitute for legal, medical or mental-health advice. If you or someone you know is in immediate danger or experiencing a mental-health crisis, contact local emergency services or a crisis line directly before relying on any system built from this document.